

PEMBROKE, ON, Canada

WMO: 711960

Lat: 45.8603N

Lon: 77.2525W

Elev: 531

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-5.9	-19.1	2.0	-12.0	-12.8	2.8	-4.8	55.6	30.1	51.7	21.9	4.6	300	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	83.0	66.8	79.3	66.0	75.8	64.4	70.5	78.0	68.5	75.8	66.6	73.2	7.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	105.1	74.9	65.9	97.5	72.3	64.0	91.2	70.2	34.8	78.4	33.1	75.8	31.5	73.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.2	40.9	35.1	-7.7	84.0	16.0	5.2	-19.3	87.8	-28.7	90.8	-37.7	93.7	-49.3	97.5
			-11.5	72.8	15.6	3.2	-22.7	75.1	-31.9	77.0	-40.7	78.8	-52.0	81.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	18.7	19.7	25.9	36.2	47.3	55.4	63.6	62.7	56.3	46.2	35.7	25.9
	DBStd	17.98	12.65	11.02	8.71	6.71	9.55	9.28	7.02	5.59	6.77	6.77	8.03	10.88
	HDD50	4563	969	849	746	420	173	50	3	0	15	160	430	748
	HDD65	8809	1434	1269	1211	865	556	309	108	108	274	584	878	1213
	CDD50	1374	0	0	0	5	89	213	426	395	203	41	2	0
	CDD65	145	0	0	0	0	7	22	66	38	12	0	0	0
	CDH74	1137	0	0	0	1	82	200	556	225	70	3	0	0
	CDH80	257	0	0	0	0	16	45	142	42	12	0	0	0
Wind	WSAvg	13.9	18.1	17.0	15.2	13.5	11.7	12.1	10.3	11.4	12.8	13.7	16.3	15.3
Precipitation	PrecAvg	30.6	2.0	1.6	1.7	2.3	2.7	2.9	3.3	3.2	3.0	2.9	2.5	2.4
	PrecMax	37.4	3.6	3.1	5.1	4.1	4.1	5.4	6.5	5.1	5.4	4.6	4.1	3.6
	PrecMin	25.2	0.7	0.6	0.5	0.7	1.3	0.7	0.8	0.8	1.4	0.5	0.8	0.7
	PrecStd	3.0	0.8	0.6	1.0	0.9	0.8	1.1	1.6	0.8	1.1	1.1	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	45.0	48.1	65.4	82.2	85.3	89.1	84.5	80.9	70.3	58.0	49.7
		MCWB	48.0	43.0	44.6	50.6	64.2	67.3	70.4	66.1	69.3	61.0	53.6	48.0
	2%	DB	41.3	39.6	43.2	56.5	75.5	79.6	84.0	79.5	75.2	64.2	54.2	44.5
		MCWB	40.0	38.1	38.7	47.1	59.8	64.8	67.6	66.6	65.8	58.6	50.9	43.4
	5%	DB	36.0	36.4	39.6	50.5	69.8	74.9	80.3	75.9	70.9	60.4	50.1	40.1
		MCWB	34.5	34.7	36.1	43.8	57.6	62.6	66.8	65.3	64.2	55.8	47.9	38.7
	10%	DB	33.0	33.8	36.8	46.5	64.0	70.5	76.4	72.5	67.0	56.6	46.5	37.2
		MCWB	31.5	32.2	34.1	40.6	54.8	61.0	65.6	64.0	61.7	53.1	44.5	35.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	42.9	45.9	57.1	67.9	71.1	74.1	71.1	71.6	64.4	55.4	48.0
		MCDB	49.0	43.7	47.6	62.6	77.1	79.4	84.7	78.5	78.7	67.4	56.7	48.8
	2%	WB	40.1	38.4	39.7	49.2	63.6	68.0	71.0	68.9	68.1	59.8	51.7	43.2
		MCDB	41.1	39.6	41.6	53.7	71.5	74.9	78.9	75.4	73.5	63.0	53.5	44.4
	5%	WB	34.8	35.1	36.7	45.2	59.5	64.7	69.0	67.1	64.8	56.3	48.1	38.8
		MCDB	35.8	36.0	38.8	50.2	66.8	71.7	76.9	73.3	69.3	59.6	50.1	39.7
	10%	WB	31.7	32.3	34.3	41.1	55.6	61.8	66.8	65.3	62.1	53.2	44.8	35.9
		MCDB	32.8	33.4	36.6	45.8	64.4	69.3	74.4	70.5	66.3	56.2	46.6	36.8
Mean Daily Temperature Range	5% DB	MDBR	13.5	13.1	12.3	12.0	15.0	15.4	16.7	14.4	14.1	11.5	10.8	10.0
		MCDBR	16.8	14.3	16.0	22.3	26.9	23.5	24.4	21.6	21.7	19.7	16.5	12.9
		MCWBR	17.1	13.4	13.1	14.7	13.0	11.1	11.4	11.0	12.6	13.9	14.6	13.6
	5% WB	MCDBR	16.7	13.5	13.9	19.4	21.4	19.5	20.4	17.6	18.4	17.2	15.1	12.3
		MCWBR	17.2	13.2	12.5	15.0	12.6	11.3	11.1	10.7	12.1	13.7	14.8	13.1
Clear-Sky Solar Irradiance	taub	0.268	0.285	0.318	0.352	0.370	0.410	0.406	0.393	0.369	0.342	0.308	0.273	
	taud	2.359	2.301	2.351	2.383	2.388	2.294	2.322	2.355	2.417	2.477	2.498	2.397	
	Ebn at Noon	268	284	289	287	285	272	272	271	269	261	252	253	
	Edh at Noon	23	30	33	36	37	41	39	36	31	26	21	20	
All-Sky Solar Radiation	RadAvg	430	731	1163	1444	1638	1804	1793	1567	1220	718	424	312	
	RadStd	60	85	102	123	146	137	149	87	115	73	48	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page